

Member	Main Area	What the Member Should Do
Member 1	Electronic Authentication Procedures + Identity Enrollment and Proofing	Explain what electronic authentication is and why it is important. Explain identity enrollment and identity proofing step by step. Show how a person, device, or system is registered and verified. Discuss possible weaknesses such as fake identities, stolen documents, deepfakes, AI-generated identities, social engineering, and weak verification. Relate the discussion to the cognitive age and explain how AI can make identity fraud more convincing. Give at least one simple real-world example.
Member 2	CSP vs Verifier vs Relying Party	Define Credential Service Provider (CSP), Verifier, and Relying Party. Explain the role of each one during authentication. Show how information moves between them during login or identity verification. Contextualize the three roles in multi-user systems, cloud environments, and hybrid clouds. Discuss vulnerabilities such as a compromised CSP, stolen credentials, weak verification, insecure communication, or a compromised relying party. Explain how AI-driven attacks can target these components. Include a simple authentication flow or diagram.
Member 3	Subscriber vs Claimant vs Applicant	Clearly define Applicant, Subscriber, and Claimant and explain how a person can move from one role to another. Give a simple example showing the difference between the three. Extend the explanation beyond human users to IoT devices, APIs, AI agents, automated systems, and agentic tools/models. Discuss vulnerabilities such as impersonation, stolen tokens, fake agents, compromised APIs, automated credential attacks, and machine-to-machine authentication problems. Explain why these risks become more serious in the cognitive age.
Member 4	Types of Authenticators + PQC + Passwordless and	Explain the main types of authenticators such as passwords, PINs, OTPs, biometrics, security keys, certificates, authenticator apps, and passkeys.

	Federated Authentication	Compare something you know, something you have, and something you are. Explain how future quantum computing may affect current cryptographic authentication and briefly introduce Post-Quantum Cryptography. Also explain passwordless authentication, passkeys, biometrics, and security keys. Cover federated authentication, SSO, OAuth, and OpenID Connect at a basic level. Discuss vulnerabilities such as biometric spoofing, token theft, compromised identity providers, account takeover, and AI-assisted phishing.
Member 5	Race-Condition Vulnerability Lab	Set up and perform the Race-Condition Vulnerability Lab. Explain what a race condition is before starting the demonstration. Show the vulnerable program or situation being tested. Follow the lab procedure and document the important commands and steps. Take screenshots showing setup, attack execution, and result. Explain why the vulnerability happens and what the attacker gains from it. Explain possible countermeasures such as secure file handling, proper permissions, atomic operations, and avoiding unsafe temporary files. Prepare a short live demonstration for the group presentation.
Member 6	Dirty-COW Attack Lab	Set up and perform the Dirty-COW Attack Lab. Explain what Dirty COW is and the idea behind copy-on-write. Show how the vulnerability can allow an attacker to modify protected or read-only data. Follow the lab procedure carefully and document the main commands and steps. Take screenshots of the environment, attack, and final result. Explain what happened during the attack and why the system was vulnerable. Discuss countermeasures such as kernel patching, system updates, limiting local access, and monitoring privilege escalation. Prepare a short live demonstration for the presentation.